

PAPER • OPEN ACCESS

CNN –based Spiral drawing classifier to detect Early-Stage Parkinson’s Disease

To cite this article: Yash Khare and Pratima Singh 2023 *J. Phys.: Conf. Ser.* **2570** 012026

View the [article online](#) for updates and enhancements.

You may also like

- [An Early Prediction of Parkinson's Disease Using Facial Emotional Recognition](#)
U Anusri, G Dhatchayani, Y Princely Angelinal et al.
- [Identifying of factor associated with parkinson's disease subtypes using random forest](#)
E Latifah, S Abdullah and S M Soemartojo
- [An End-to-End Deep Learning Approach for Diagnosis of Parkinson's Disease Based on Hand Drawing](#)
Yifan Wang and Yifan Zhang

CNN –based Spiral drawing classifier to detect Early-Stage Parkinson’s Disease

Yash Khare* and Pratima Singh

Ajay Kumar Garg Engineering College, Ghaziabad, Uttar Pradesh, India

yash2010118@akgec.ac.in

Abstract. Parkinson's disease is a neurological condition wherein one experiences a loss in control of motor movements. The disease usually infects gradually but worsens over time. Parkinson's often starts with a tremor in one hand. Slow movements, stiffness in the body, and loss of balance are some of the major symptoms of the disease. Damaged nerve cells present in the brain cause drops in the levels of dopamine in the body. It is very difficult to detect Early-Stage Parkinson's Disease.

Researchers have now been able to detect Parkinson's disease at an early stage (Perhaps at Stage 2 or Stage 3), but there have been no significant advancements in the detection of Parkinson's at the initial stage. Analyzing the handwriting styles can be one of the initial symptoms of the disease. The dataset given by NIATS of the Federal University of Uberlandia was used in our research. In this study, we present a CNN model that can distinguish between spiral drawings made by patients with Parkinson's disease and those made by people not suffering from Parkinson's disease with high accuracy.

Keywords – CNN, optimization, Parkinson's disease

1. Introduction

After Alzheimer's disease, Parkinson's disease is the most prevalent neurological disease, it impacts 2% to 3% of the individuals ≥ 60 -65. The neuro and pathological markers of Parkinson's disease include demyelination in the substantia-nigra, which produces striatal dopamine decrease in the body, and intracellular inclusions harboring clumps of α -synuclein [6]. The inability to regenerate neurons is the primary cause of Parkinson's disease. Neurons begin to die off as a person ages and are irreplaceable. Dopamine is a chemical fluid secreted by neurons that is entirely responsible for movement in the body and signal transmission between neurons. The neurological disease slows down as Dopamine levels diminish with age, regulated by the numerous communication channels in the brain [7,9].

Tremor, stiffening of bones, bradykinesia/akinesia, and problems in maintaining the body posture are the premier motor symptoms of Parkinson's disease, but the picture also includes various motor and non-motor symptoms. Many drugs have been developed to reduce the spread of the disease, but their usage over the years might catalyze the process of neuro degeneration. The main problem regarding PD concerns its detection in the early stages. This poses a problem to date [8].

Although no specific test exists to diagnose PD, there are a few physical tests, usually associated with other neurological disorders which can be used to predict Parkinson's disease. Handwriting abnormalities are one of the most common symptoms in people with Parkinson's disease (PD). The computer-assisted analysis of handwriting enables to study some of the potential patterns that may be beneficial in the detection and classification of Parkinson's disease. This study aims at predicting



Parkinson's disease using the "Famous Spiral Test". We collected some images of spirals drawn by healthy people and those having Parkinson's and trained our Convolutional Neural Network model to study and differentiate features and predict future predictions. Work has been done in this field with the target to detect Spiral and Wave Drawings for Early-Stage Parkinson's disease [8] with an accuracy of 87%. We aim to increase the accuracy of the model. We selected the best model by tuning various filters and selecting the best filter at each step for future computation. We thus, propose a CNN model with 92.58% accuracy.

2. Background

Parkinson's disease (PD), sometimes known as Parkinson's, is a degenerative illness of the central nervous system that predominantly affects the motor behaviour of a person. The symptoms take time to appear, and as the disease progresses through the phases, non-motor symptoms become more prevalent. Tremors, stiffness, slowness of movement, and troubles in walking are common early signs. Depression, anxiety, and apathy are all possible symptoms. People may also suffer trouble sleeping.

The condition is caused by the decrease in no of the cells in a portion in the mid brain area called the substantia nigra, resulting in a dopamine deficiency. The reason for death of cells remains unclear, although it involves the synthesis of misfolded-proteins to Lewy bodies in neurons.

The aetiology of Parkinson's disease is uncertain. Because some genes are known to be inheritable, already infected member of the family are at a higher risk of having it. People with past history of prolonged illness or head injuries or have been exposed to several toxic chemicals are at huge risk of developing Parkinson's disease. People who smoke tobacco and drink hot beverages like tea and coffee, all have an increased chance of developing Parkinson's disease [10].

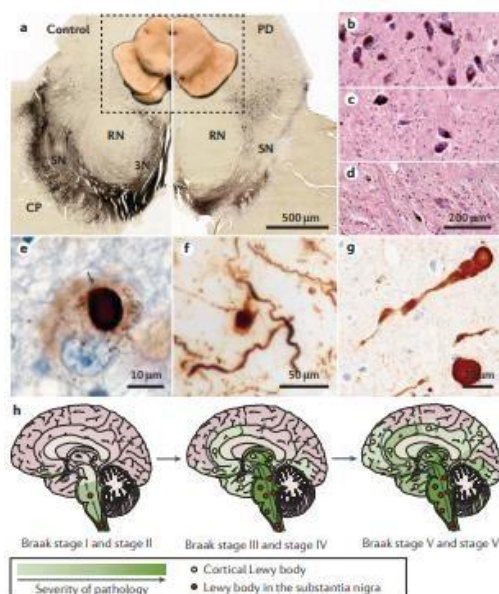


Figure 1. The main neuropathologies of Parkinson's disease [2]

To rule out other illnesses, neuroimaging (magnetic resonance imaging to look for dopamine neuronal dysfunction) might be used. Parkinson's disease can affect about 1% of the population and is most frequent in people over the age of 70. Males are threefold higher as likely to be affected as females. In 2015, PD struck 6.2 million individuals globally, killing roughly 117,400 people. Following a

diagnosis, the usual average lifespan is 7 to 15 years [3].

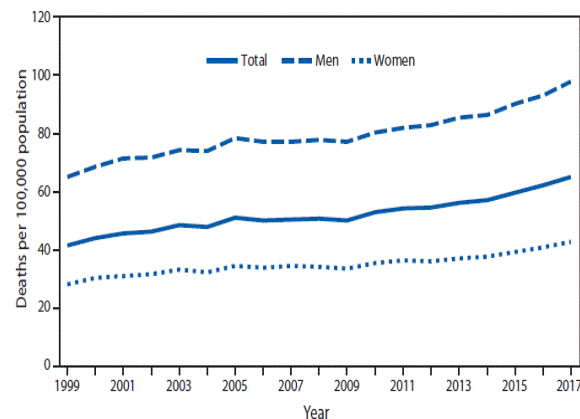


Figure 2. Parkinson's Disease Death Rates among Adults Aged 65 Years and greater [4]

While there exists no solution to Parkinson's disease, this treatment aims to lessen the severity of the symptoms. First-line therapy includes levodopa (L-DOPA), MAO-B inhibitors, and dopamine agonists. As the illness worsens, these medications become less effectual, and as a side effect, they cause uncontrolled muscle movements. At that time, medication combinations and doses may be used. Diet and other forms of rehabilitation have been demonstrated to help in symptom relief. A surgical approach to embedding electrodes of micro-size for deep mind activation has been used in severe cases when medical medicines have failed to treat motor symptoms. Evidence for treatments for non-movement-associated signs and symptoms of Parkinson's disease, together with sleep issues and emotional issues, remains being investigated[5][18].

When motor issues affect the dominant hand, patients might express poor handwriting among the first symptoms. The existence of any of the premier motor symptoms in people with Parkinson's disease can dramatically affect the quality of their handwriting. Handwriting is a difficult task that demands cognitive, perceptual, and fine motor skills. Handwriting impairment is one of the first indicators that lead people to seek medical help, even though it is not explicitly included in the diagnostic criteria for Parkinson's disease.

Handwriting kinematics changes are one of the most recently proposed indicators for Parkinson's disease. According to a recent study, handwriting analysis has the potential to be used for both early diagnosis and monitoring illness development. Handwriting has a unique function to play in the evaluation of neurodegenerative diseases. Learning and practicing handwriting involves the cerebral cortex, basal ganglia, and cerebellum. It is a complex activity with cognitive, kinesthetic, and perceptual-motor components. Handwriting issues can be linked to both the disease and its severity, therefore alterations in writing can be used as a biomarker. Micro-graphia (a significant reduction in size of letters and alphabets) is, for example, a well-known symptom of Parkinson's disease that may be easily diagnosed using traditional pen-and-paper activities[4][19-21].

Handwriting tasks in PD are usually identified by:

- lesser control to motor movement control, i.e., Letter size decrease(micrographia) and failure to uphold characters' stroke size as the person continues to write further [11] [12].
- changes in active aspects, such as increased motor time, decreased velocities and accelerations, and a corresponding rise in rate of writing texts (irregular and brady-kinetic movements) [13] [14].

To build differential diagnoses and follow the disease's course, it's necessary to assess the recovery of symptoms related to movement in response to dopaminergic medications. Although

improvements in symptoms related to movement with dopaminergic treatment are one of the features of PD, the significance of improvement in fine motor skills with this medication is uncertain. On the other hand, recent advancements in digitizing tablet technology have changed the study of different elements of handwriting. These technologies enable the objective evaluation of handwriting fluency and velocity and as well as the quantification of letter size, which is hard to do with paper and pen. An objective examination of a variety of handwriting-related metrics might shed light on the role of fine motor skills in Parkinson's disease [15][23-26].

3. Survey

After doing extensive research, we discovered various study articles with comparable aims to ours but distinct methodologies. We researched a few ways to solve the classification including the use of Classification algorithms at the beginning. After a brief review, we collected some data samples –

Table 1. Models based on deep learning to predict Parkinson's disease [15]

Attributes worked upon	Classification algo used	Dataset used	Accuracy
Active Attributes	Support Vector Machine	Data based on handwriting by Parkinson's affected patients	80.09
Active Attributes	SVM	PAHAW	79.4
Active Attributes	SVM	PAHAW	78.84
Active Attributes and spatial and temporal Attributes	SupportVM classification with Radial-Gaussian - Kernel (RGK)	PAHAW	88.13
Active Attributes	SupportVM with radial basis function kernel	PAHAW	89
Active and constraint Attributes	SupportVM, Ada-Boost, K-Nearest Neighbors	PAHAW	81, 79, 72
Not Automated Attributes	Naive Bayes, Optimum path forest, SVM	HPD	79, 77, 76
Time based Attributes	CNN	HPD	93.50
Attributes based on pen drawings	CNN	HPD	80.19

Table 2. Summary of various motion tasks performed to predict Parkinson's disease[8]

Exercise	Attributes	Classification algorithm used	Accuracy
Transcription and location copying	On-surface + in-air attributes	Discriminant Analysis	97.15
Depictions of Spiral and Meander	Grip and tilt pressure	Convolutional Neural Networks	87.14
Depictions of lines	Position, Velocity, Mean Vel., Entropy	Naive Bayes	88.63
Transcription of Sentences and other basic constructs	Displacement pressure, average speed rate, max acceleration	Naive Bayes	93.3

We then settled upon Convolutional Neural Networks because it requires very minimal pre-processing in comparison to other image classification methods. This implies that the network learns to choose the right filters (or kernels) by self-learning (i.e. Learning while in the training process), whereas in traditional methods, these filters are hand-engineered. This feature extraction independence from past information and human interaction is a significant benefit. CNNs are fully connected feed forward neural networks. CNNs are very effective in **reducing the number of parameters without losing on the quality of models**. Since we wanted to extract features from those suffering Parkinson's and those who are not, we considered CNN as the best model to train on our dataset. **The above-mentioned paper provides a Convolutional Neural Network solution with an accuracy of 87.28%**. Our experiment aimed at increasing the accuracy of the CNN model.

We trained a CNN model on a training dataset consisting of 500 images of spiral drawings made by healthy people and 500 images of spiral drawings made by people suffering from Parkinson's disease. We then tested our model on a test data set consisting of 125 images of spiral drawings made by people suffering from healthy as well as 125 images of spiral drawings made by people suffering from Parkinson's disease. The dataset was provided by A.O. Andrade and J.P. Folado from the NIATS of the Federal University of Uberlandia.

We tweaked our model's training by changing the following parameters while selecting the best accuracy result from previous parameters encountered-

1. Change of activation functions
2. Change of No. of strides
3. Change of no. of filters
4. Change of optimizer
5. Change of batch and kernel size

Table No.3 shows the results of the experiments conducted while adjusting the parameters:

Table 3. Variation of the accuracy of change of parameters

S.No.	Parameters	Options	Accuracy
1.	Change of Activation functions in hidden layers	RELU activation function	49.56%
		SELU activation function	71.22%
		ELU activation function	84.28%

		TanH activation function	60.71%
		Softmax activation function	50.00%
2.	Change of No. of Strides (with ELU activation function)	1	80.11%
		2	84.28%
		3	80.56%
		4	66.50%
3.	Change of No. of filters(with ELU and 2 strides)	32	68.15%
		64	85.33%
		128	77.44%
4.	Change of Optimizer(with ELU and 2 strides and 64 filters)	Adam	88.46%
		SGD	74.55%
		Adagrad	48.24%
		RMSprop	61.41%
		Nadam	78.28%
5.	Change of Batch Size(with ELU and 2 strides and 64 filters and Adamoptimizer)	15	64.55%
		5	89.00%
		4	92.58%
		3	81.94%
		2	83.48%
6.	Change of Kernel Size(with ELU and 2 strides and 64 filters and Adamoptimizer and Batch size = 4)	3	72.70%
		5	92.58%
		6	82.88%
		7	80.56%
7.	Change of Target Size(with ELU and 2 strides and 64 filters and Adamoptimizer and Batch size = 4 and Kernel Size = 5)	32,32	92.58%
		64,64	77.78%
		128,128	48.88%

4. Results and Conclusion

A CNN model considering the following architecture was developed, due to its maximum efficiency. The number of hidden layers in the model is providing the best results. The basic architecture of the proposed model is as mentioned in the diagram:

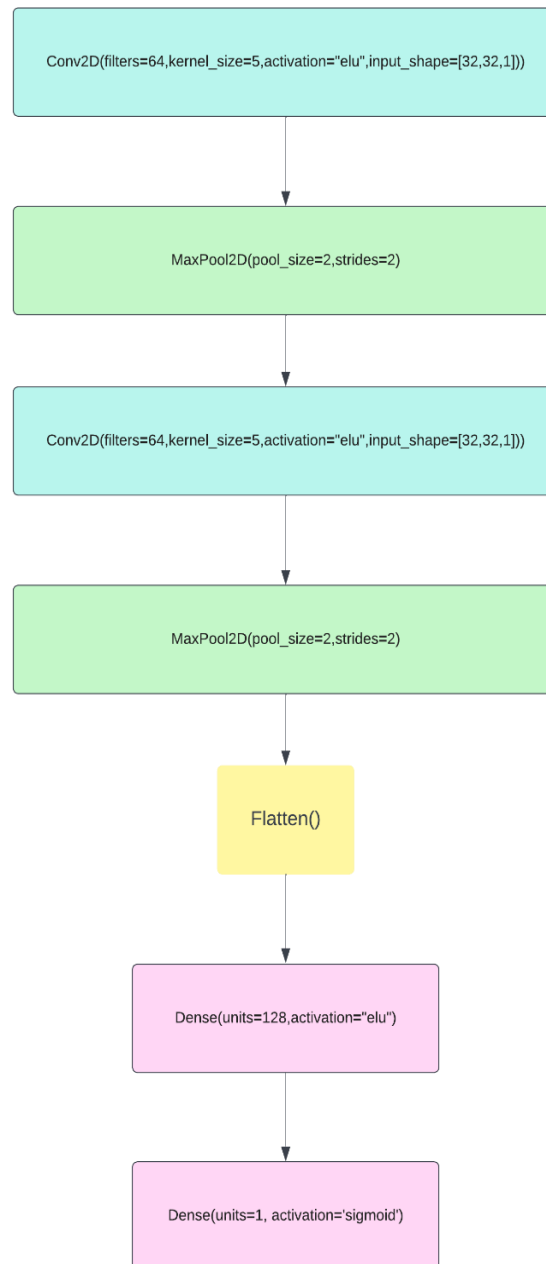


Figure 3. Architecture of the proposed CNN model

The following change of parameters were observed while choosing the best result from the previous parameter and feeding it into the successive parameters. The steps followed were -

1. Change of the activation function in the convolutional layers ranging from RELU, SELU, ELU, TanH, and softmax function. We achieved max accuracy of 84.28% with the ELU function.
2. Change of the no. of strides in the pooling layers ranging from 1 to 4 with max accuracy of 84.28% with 2 strides (while keeping ELU as activation function)
3. Change of the no. of filters while keeping ELU as activation function and 2 strides ranging from 32 to 128 with max accuracy of 85.33% in 64 filters
4. Worked upon the optimizers available for the model ranging from Adam to SGD to Adagrad to RMSprop to Nadam with Adam as the best optimizer with max accuracy of 88.46% while keeping the previous parameters with max efficiency

5. Change of batch size from 1 to 15 gave batch size = 4 with max accuracy of 92.58%
6. Change of Kernel size in the convolutional layer from 3 to 7 gave max accuracy of 92.58% with kernel size = 5
7. Change of Target size while predicting from (32,32) to (128,128) gave max accuracy of 92.588% with (32,32) target size while keeping the previous parameters with max efficiency

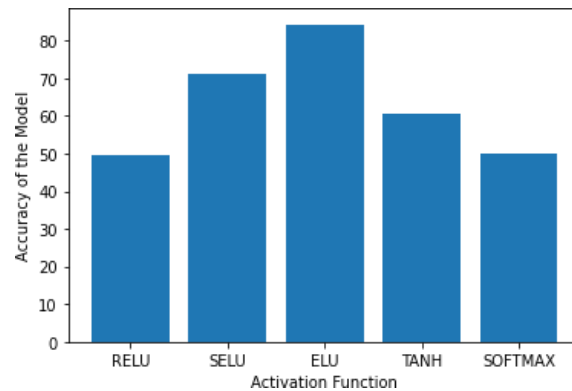


Figure 4. Change of Accuracy upon change of activation function

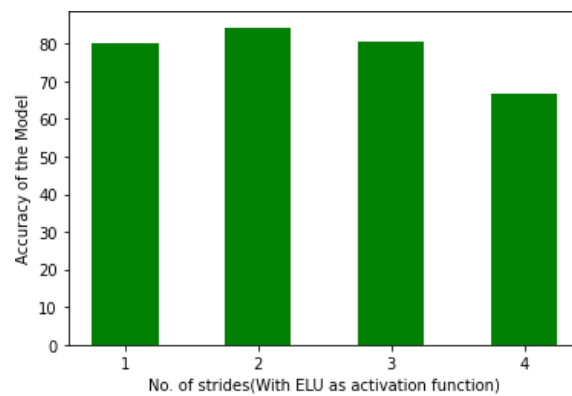


Figure 5. Change of Accuracy upon change of no. of strides

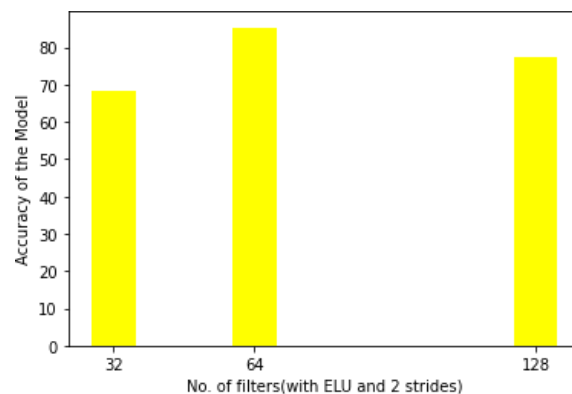


Figure 6. Change of Accuracy upon change of no. of filters

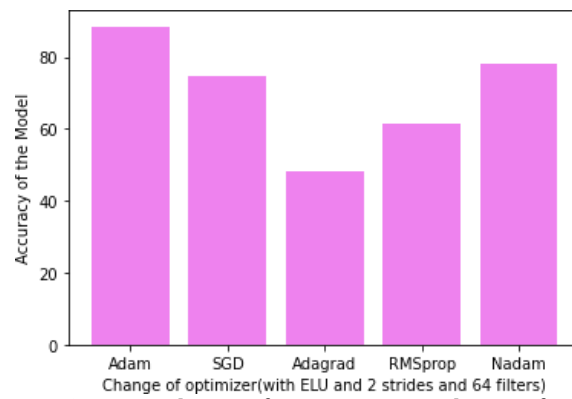


Figure 7. Change of Accuracy upon change of optimize

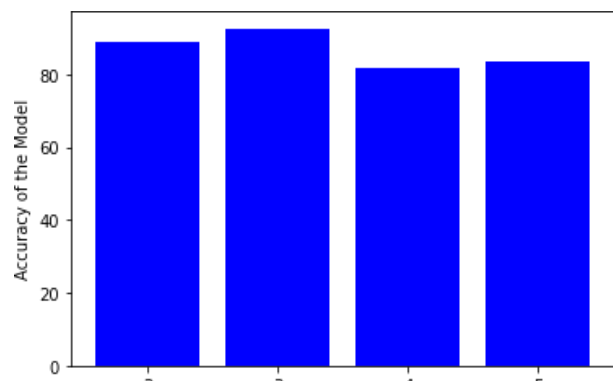


Figure 8. Change of Accuracy upon change of batch size

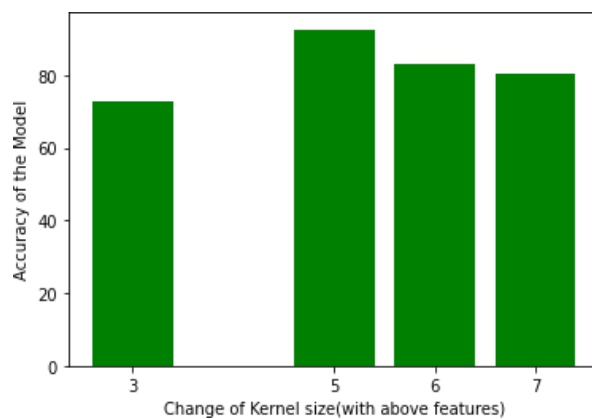


Figure 9. Change of Accuracy upon change of kernel size

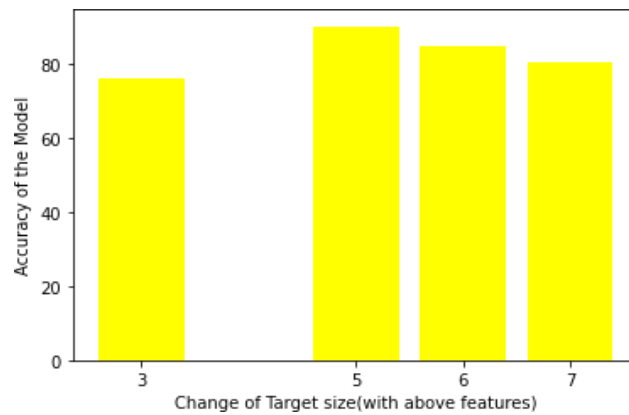


Figure 10. Change of Accuracy upon change of Target Size

Finally, a CNN model was developed with an accuracy of 92.58% with the following parameters -

1. ELU activation function
2. No. of strides = 2
3. No. of filters = 64
4. Adam optimizer
5. Batch size = 5
6. Kernel size = 5
7. Target size = (32,32)

The previous research in the field stated an accuracy of 87% and this research improves the accuracy of the model by reaching an accuracy of 92.58%.

This paper opens opportunities for researchers to research upon the field and to work upon models to increase accuracy to achieve better results. Although this paper states accuracy of 92.58% there is a huge scope of improvement on the same.

5. References

- [1] Kayalibay, B., Jensen, G., & van der Smagt, P. (2017). CNN-based segmentation of medical imaging data. *arXiv preprint arXiv:1701.03056*.
- [2] Arena, P., Basile, A., Bucolo, M., & Fortuna, L. (2003). Image processing for medical diagnosis using CNN.
- [3] Li, Q., Cai, W., Wang, X., Zhou, Y., Feng, D. D., & Chen, M. (2014, December) medical image classification with convolutional neural networks. *IEEE*
- [4] Bullock, J., Cuesta-Lázaro, C., & Quera-Bofarull, A. (2019, March). XNet : A convolutional neural networks implementation for medical X-ray image segmentation suitable for small datasets. In *Molecular Imaging 2019: Biomedical Applications in Molecular, Structural and Functional Imaging* (Vol. 10953, pp. 453-463). SPIE
- [5] Zhang, Y., Lobo-Mueller, E. M., Karanicolas, P., Gallinger, S., Haider, M. A., & Khalvati, F. (2020). CNN-based survival model for pancreatic ductal adenocarcinoma in medical imaging. *BMC medical imaging*, 20
- [6] Werner Poewe Klaus Seppi and 6 others' article on "Parkinson's Disease," Article number: 17013 (2017) published in *Nature Reviews Disease Primers* -
- [7] Clayton R. Pereira, Danilo R. Pereira, and 5 others' paper on "Handwritten dynamics assessment through convolutional neural networks published in Volume 87 of "Artificial Intelligence in Medicine"
- [8] D. Impedovo and G. Pirlo, "Dynamic Handwriting Analysis for the Assessment of Neurodegenerative Diseases: A Pattern Recognition Perspective," in *IEEE Reviews in Biomedical Engineering*, vol. 12, pp. 209-220, 2019, DOI: 10.1109/RBME.2018.2840679.
- [9] da Silva, F. C., Iop, R. D. R., de Oliveira, L. C., Boll, A. M., de Alvarenga, J. G. S., Gutierrez Filho, P. J. B., & da Silva, R. (2018). Effects of physical exercise programs on cognitive function in Parkinson's disease patients: a systematic review of randomized controlled trials of the last 10 years. *PloS one*, 13(2), e0193113.
- [10] <https://ncbi.nlm.nih.gov/books/NBK536726>
- [11] Teulings, H. L., Contreras-Vidal, J. L., Stelmach, G. E., & Adler, C. H. (2002). Adaptation of

- handwriting size under distorted visual feedback in patients with Parkinson's disease and elderly and young controls. *Journal of Neurology, Neurosurgery & Psychiatry*, 72(3), 315-324.
- [12] Martins, R., Oliveira, F., Moreira, F., Moreira, A. P., Abrunhosa, A., Januário, C., & Castelo-Branco, M. (2021). Automatic classification of idiopathic Parkinson's disease and atypical Parkinsonian syndromes combining [11C] raclopride PET uptake and MRI grey matter morphometry. *Journal of neural engineering*, 18(4), 046037.
- [13] Van Gemmert, A.W., Teulings, H.-L., Stelmach, G.E.: Parkinsonian patients reduce their stroke size with increased processing demands. *Brain Cogn.* 47(3), 504–512 (2001)
- [14] Drotar, P., Mekyska, J., Smekal, Z., Rektorova, I., Masarova, L., Faundez-Zanuy, M.: Prediction potential of different handwriting tasks for diagnosis of parkinson's. In: 2013 EHealth and Bioengineering Conference (EHB), pp. 1–4. IEEE (2013)
- [15] Nutt, J.G., Wooten, G.F.: Diagnosis and initial management of parkinson's disease. *N. Engl. J. Med.* 353(10), 1021–1027 (2005)
- [16] Kamran, I., Naz, S., Razzak, I., & Imran, M. (2021). Handwriting dynamics assessment using deep neural network for early identification of Parkinson's disease. *Future Generation Computer Systems*, 117, 234-244.
- [17] Alakananda Chakraborty *et al* 2021 *J. Phys.: Conf. er.* **1854** 012040
- [18] Jindal, M., Bajal, E., Chakraborty, A., Singh, P., Diwakar, M., & Kumar, N. (2020). A novel multi-focus image fusion paradigm: A hybrid approach. *Materials Science*, 2214, 7853.
- [19] Dabeer, S., Khan, M. M., & Islam, S. (2019). Cancer diagnosis in histopathological image: CNN based approach. *Informatics in Medicine Unlocked*, 16, 100231. Patra, A. K., Ray, R., Abdullah, A. A., & Dash, S.
- [20] R. (2019, November). Prediction of Parkinson's disease using Ensemble Machine Learning classification from acoustic analysis. In *Journal of physics: conference series* (Vol. 1372, No. 1, p. 012041). IOP Publishing.
- [21] Lee, H., & Kwon, H. (2017) Going deeper with contextual CNN. for hyperspectral image classification. *IEEE Transactions on Image Processing*, 26(10), 4843-4855.
- [22] Gil-Martín, M., Montero, J. M., & San-Segundo, R. (2019). Parkinson's disease detection from drawing movements using convolutional neural networks. *Electronics*, 8(8), 907.
- [23] Gorzelic, P., Schiff, S. J., & Sinha, A. (2013). Model-based rational feedback controller design for closed-loop deep brain stimulation of Parkinson's disease. *Journal of neural engineering*, 10(2), 026016.
- [24] Xiao, B., He, N., Wang, Q., Cheng, Z., Jiao, Y., Haacke, E. M., & Shi, F. (2019). Quantitative susceptibility mapping based hybrid feature extraction for diagnosis of Parkinson's disease. *NeuroImage: Clinical*, 24, 102070.